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Mini Project Synopsis

On

“Automatic rain shelter”

Submitted in partial fulfilment for the award of degree of
Bachelor of Engineering

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AUTOMATIC RAIN SHELTER

AIM: The aim of the project is to develop a automatic rain shelter.

Abstract:

In this proposal system presents the design and development of an automatic rain shelter system aimed at enhancing user convenience and protection from adverse weather conditions. The system employs a network of sensors detect rainfall and deploys a collapsible shelter mechanism in real-time. Utilizing advanced materials for lightweight durability. User-friendly interfaces allow for manual control. The system demonstrated rapid deployment, effective weather protection, and user satisfaction. This innovation has potential applications outdoor events, public spaces, and personal use.

INTRODUCTION

As urban environments continue to evolve, outdoor activities and events are becoming increasingly popular. However, unexpected weather changes, particularly rain, can disrupt these experiences and pose challenges for participants. The need for effective and convenient weather protection solutions has never been more critical. This proposal introduces an innovative automatic rain shelter designed to provide quick and reliable coverage in response to sudden rainfall. Leveraging advancements in sensor technology and automated systems, this rain shelter offers a seamless user experience by automatically deploying when rain is detected. The integration of lightweight, durable materials ensures that the shelter is both portable and robust. This solution not only addresses the inconvenience of inclement weather but also aims to improve the enjoyment of outdoor activities. In this proposal system, we explore the system's design, functionality, and real-world applications, highlighting its potential to transform how we interact with outdoor spaces in varying weather conditions. By bridging the gap between technology and everyday needs, the automatic rain shelter stands as a testament to innovation in environmental adaptability.

LITERATURE SURVEY

1. Automatic Roof for Crop:

Agriculture is the spine of our country. over it, nearly 70% of profit for our country are coming from agriculture. Nowadays, in the course of the wet seasons, the harvested plants get affected because of the heavy rainfall. The main goal of this task seems to be to protect the seedlings against intense or unusual rains. For automated rooftop construction, a rain detector is employed. This gadget entails protecting the seeds with the aid of using the automobile cover which can cover the complete place in which the crop is spread.

Published in: [2023 IEEE 3rd International Conference on Technology, Engineering, Management for Societal impact using Marketing, Entrepreneurship and Talent \(TEMSMET\)](#)

Date of Conference: 10-11 February 2023

Date Added to IEEE Xplore: 16 June 2023

Author :- T Jaswanth Yadav, E. Naga Harini, B. Manoj Shankar.

2. Full Protection Covers for Parked Car at Remote Stations:

The aim of this paper is to give a review on the car covers from distant areas to provide protection to car from rain and hailing. Here, we mentioned many technologies used till now for protection and how we have taken an automation system to protect car by using IOT based system attached with a mobile application which will operate the whole system. The sheet will cover the whole car within seconds after having the command from remote areas. The cover is attached with the car itself. Construction of this automation has been described as the combined use of mechanical and electronic systems to achieve automatic operation or control to reduce problems. This paper briefly describes the problem observed during night when people forgot to cover their cars and how our device would help them to solve these problems.

Published in: [2019 5th International Conference on Signal Processing, Computing and Control \(ISPCC\)](#)

Date of Conference: 10-12 October 2019

Date Added to IEEE Xplore: 10 February 2020

Author :- Kirti Masown, Rajesh Mehra

3.Design of Raindrop detector based on Arduino

The Internet of Things technology and sensor technology can be used to analyze and process the rainy day data, and the raindrop detector based on Arduino is built. Arduino is a convenient, flexible and easy-to-use open source electronic platform, including various hardware and software versions. Arduino is built in an open source panel and USES a development environment similar to Java and C language. It mainly consists of two parts. The first part is the hardware part, that is, the designed hardware circuit is connected to the Arduino board, and the other part is the Arduino IDE. The sensors used include water level depth sensor, large area raindrop sensor and temperature and humidity dht11 sensor. By connecting these sensors to the Arduino board, and then writing the code of the sensor and the code connected to the mobile phone network into the board, the monitoring of rainy weather can be realized.

Published in: [2021 6th International Conference on Intelligent Computing and Signal Processing \(ICSP\)](#)

Date of Conference: 09-11 April 2021

Date Added to IEEE Xplore: 26 April 2021

Author :- Li Ru, Zhai Shu-ying

4.Development of a Raspberry Pi Based Rain Predictor for Smart Retractable Roof Systems using Convolutional Neural Networks: A Study Based on Daylight Conditions:

Technological advancements have made it possible to develop intelligent retractable roofs. These are used widely in home balconies, soccer stadiums, and smart greenhouses, to mention a few. However, the current retractable roof systems are fitted with rain sensors that need to be in direct contact with the rain droplets before they can send a signal to the control circuit to close the retractable roof. With such sensors, when the rainfall rate is high, they may be disturbing to property supposedly protected by the retractable roof when the rain is detected. For instance, nurturing of the crops in greenhouses is controlled by adhering to the required amounts of the soil nutrients and water concentration. So, the current sensors for rain detection that are installed in smart retractable roof systems for greenhouses may allow excess water concentrations than required as they need to be in direct contact with the rain droplets before the roof can close. The purpose of this study is to develop a Raspberry Pi camera-based system using Convolutional Neural Networks (CNN) to predict the possibility of rainfall by examining

the cloud condition. The advantage of this method is that the rain is detected early before it falls hence preventing monitored environments from rainfall damages. The custom CNN model was trained on images of the clouds with two classes. The first class was formed by images of clear clouds while the second class was formed by grey to dark clouds that form before the rainfall. The model's performance was improved by fine tuning it's learning to a performance that reduced misclassifications while increasing the classification accuracy from 84% (at a learning rate of 0.01) to 97% (at a learning rate of 0.00001).

Published in: [2021 International Conference on Artificial Intelligence, Big Data, Computing and Data Communication Systems \(icABCD\)](#)

Date of Conference: 05-06 August 2021

Date Added to IEEE Xplore: 30 August 2021

Author:-Malusi Sibiya, Bernardus C.

OBJECTIVES

1. To develop a robust system that utilizes advanced sensors to detect rainfall and environmental changes in real-time, enabling timely shelter deployment.
2. Incorporate features based on user feedback, such as customizable settings for deployment sensitivity and size, to enhance user satisfaction and usability.
3. Implement safety measures to ensure the shelter remains stable during adverse weather conditions and does not pose risks to users or surrounding structures.

METHODOLOGY

Developing a basic model of an automatic rain shelter involves a streamlined approach focused on essential features while ensuring functionality and user convenience. The following methodology outlines the steps required to design, build, and test a basic prototype.

1. Research and Concept Development

Objective: Understand the requirements and existing solutions.

Activities: Conduct a literature review on existing rain shelters and automatic deployment systems.

2. Design Planning

Objective: Create initial design specifications.

Activities: Sketch design concepts focusing on a collapsible structure that is easy to deploy.

3. Component Selection

Objective: Choose essential components for the shelter.

Activities: Select basic sensors (e.g., a rain sensor) to detect rainfall. Choose a simple motor mechanism for deploying and retracting the shelter. Identify a power source, such as a rechargeable battery.

4. Prototype Development

Objective: Build a functional prototype.

Activities: Construct the frame using lightweight materials to ensure portability.

5. Control System Setup

Objective: Develop a basic control system.

Activities: Use a microcontroller (like Arduino or Raspberry Pi) to manage sensor input and control the motor. Write a program to deploy the shelter when rain is detected and retract it once the rain stops.

6. Final Review and Documentation

Objective: Prepare for potential future developments.

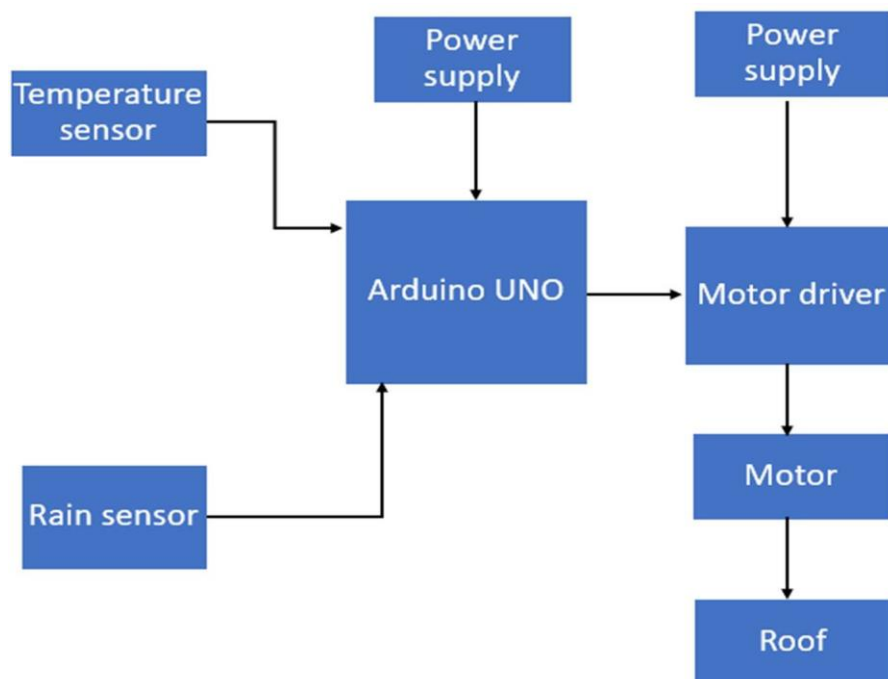
Activities: Document the design process, materials used, and results from testing.

HARDWARE REQUIREMENT

- 1.Arduino uno
- 2.Motor Driver
- 3.Motor
- 4.Rain Detecting Sensor
- 5.Battery
- 6.Jumper Wires

SOFTWARE REQUIRED:-

- 1.Arduino IDE



ADVANTAGES

1. Convenience: Automatically deploys when rain is detected, providing instant protection without requiring user intervention. This is particularly useful during unexpected weather changes.
2. Enhanced Outdoor Experience: Encourages outdoor activities by ensuring protection from rain, making it suitable for events, picnics, and public spaces.
3. Time-Saving: Eliminates the need to manually set up a shelter, allowing users to focus on enjoying their time outdoors rather than worrying about weather conditions.
4. Durability: Constructed from weather-resistant materials, ensuring longevity and reliability even in harsh conditions.

APPLICATIONS

1. Outdoor Events: Ideal for concerts, festivals, and fairs, providing instant protection for attendees and equipment from unexpected rain.
2. Public Parks and Recreation Areas: Offers shelter for visitors during picnics, sports activities, and leisure time, enhancing the overall outdoor experience.
3. Domestic Use: Serves as a personal rain shelter for gardens, patios, or balconies, allowing homeowners to enjoy outdoor spaces without worrying about the weather.
4. Sports Facilities: Protects players and spectators during outdoor sports events, ensuring that games can continue or that fans remain dry.
5. Construction Sites: Protects workers and equipment from rain, allowing construction activities to continue with minimal disruption.

EXPECTED OUTCOMES

1. Increased Outdoor Engagement: By providing reliable protection from rain, more people will be encouraged to participate in outdoor activities, events, and gatherings.
2. Improved Weather Adaptability: The shelter will demonstrate effective real-time response to weather changes, ensuring users stay dry and comfortable during sudden downpours.
3. User Satisfaction: Positive feedback is expected regarding the intuitive controls and mobile app integration, leading to high levels of user satisfaction and repeat usage.
4. Durability and Longevity: The shelter is anticipated to withstand various weather conditions, showcasing its durability and reliability over time, which reduces maintenance costs.

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